



Exercise 3: Job Scheduling Algorithm Simulation

Lappeenranta–Lahti University of Technology LUT

Computational Science and Artificial Intelligence

CT30A3350 Operating Systems and System Programming (Lahti)

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Task 1

The job information was first read from a text file and then printed out to check if everything was correct. The output matched the given table, which means that the input data had been loaded properly.

```
hamidur@DESKTOP-7CGRQ8S:~/LUT_exercises/os_sp/ex3$ gcc -o job_schedule job_schedule.c
hamidur@DESKTOP-7CGRQ8S:~/LUT_exercises/os_sp/ex3$ ./job_schedule
please input job data file name
jobs.txt

Original job data
-----
jobID  arrival  need   priority
1      800      50     0
2      815      30     1
3      830      25     2
4      835      20     2
5      845      15     2
6      700      10     1
7      820      5      0
```

Figure 1. Terminal showing the job information printed correctly

Task 2

```
FCFS result
-----
```

jobID	arrival	need	start	wait	turnaround
6	700	10	700	0	10
1	800	50	800	0	50
2	815	30	850	35	65
7	820	5	880	60	65
3	830	25	885	55	80
4	835	20	910	75	95
5	845	15	930	85	100

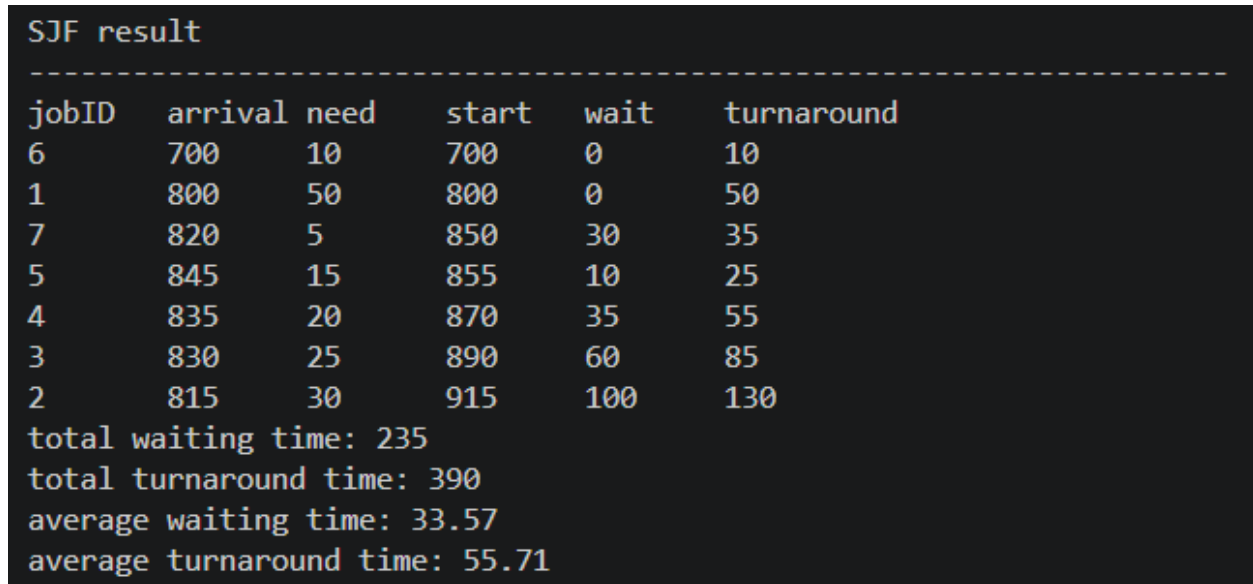
```
total waiting time: 310
total turnaround time: 465
average waiting time: 44.29
average turnaround time: 66.43
```

Figure 2. Terminal showing the resulting output of FCFS

In FCFS scheduling, the jobs were executed based on their arrival time. The execution order was 6, 1, 2, 7, 3, 4 and 5. The total waiting time was 310, and the total turnaround time was 465. The average waiting time was 44.29 and an average turnaround time was 66.43.

FCFS is easy to understand and follows a fair first come first served rule. But it can increase waiting time because short jobs that arrive later still have to wait behind longer jobs that arrived earlier.

Task 3

A terminal window with a dark background and light-colored text. The title 'SJF result' is at the top, followed by a dashed line. Below is a table with 6 columns: jobID, arrival, need, start, wait, and turnaround. The rows show the execution order of jobs 6, 1, 7, 5, 4, 3, and 2. At the bottom, four summary statistics are listed: total waiting time, total turnaround time, average waiting time, and average turnaround time.

```
SJF result
-----
jobID  arrival  need   start   wait   turnaround
6      700     10    700     0      10
1      800     50    800     0      50
7      820     5     850    30     35
5      845    15    855    10     25
4      835    20    870    35     55
3      830    25    890    60     85
2      815    30    915   100    130
total waiting time: 235
total turnaround time: 390
average waiting time: 33.57
average turnaround time: 55.71
```

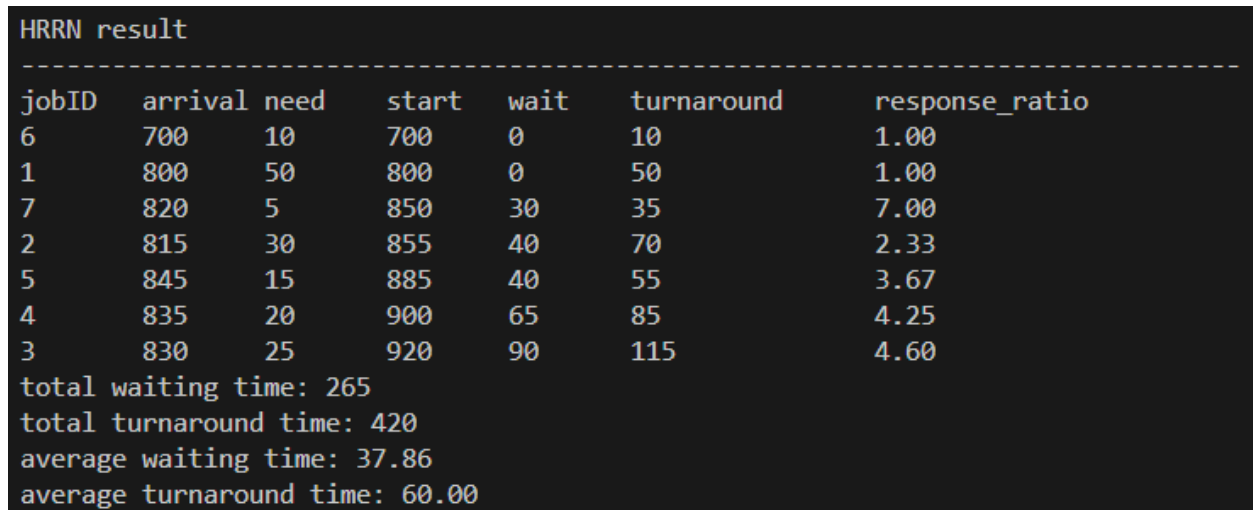
Figure 3. Terminal showing the resulting output of SJF

In SJF scheduling, the program selected the shortest job among the ones that had already arrived. The execution order was 6, 1, 7, 5, 4, 3 and 2. The total waiting time was 235 and the total turnaround time was 390. The average waiting time was 33.57 and an average turnaround time was 55.71.

Compared to FCFS, SJF gave better results by reducing both waiting time and turnaround time. This happened because the shorter available jobs were finished earlier instead of waiting behind longer ones.

Task 4

Highest Response Ratio Next Algorithm

A terminal window with a dark background and light-colored text. The title 'HRRN result' is at the top. Below it is a table with 7 columns: jobID, arrival, need, start, wait, turnaround, and response_ratio. The table contains 7 rows of data for jobs 6, 1, 7, 2, 5, 4, and 3. Below the table, four summary statistics are listed: total waiting time: 265, total turnaround time: 420, average waiting time: 37.86, and average turnaround time: 60.00.

jobID	arrival	need	start	wait	turnaround	response_ratio
6	700	10	700	0	10	1.00
1	800	50	800	0	50	1.00
7	820	5	850	30	35	7.00
2	815	30	855	40	70	2.33
5	845	15	885	40	55	3.67
4	835	20	900	65	85	4.25
3	830	25	920	90	115	4.60

total waiting time: 265
total turnaround time: 420
average waiting time: 37.86
average turnaround time: 60.00

Figure 4.1. Terminal showing the resulting output of HRRN

In HRRN scheduling, the job with the highest response ratio was chosen at each step. The execution order came out as 6, 1, 7, 2, 5, 4 and 3. The total waiting time was 265 and the total turnaround time was 420. So, the average waiting time was 37.86 and the average turnaround time was 60.00.

HRRN gives a balance between short jobs and fairness. It still favors shorter jobs, but jobs that have been waiting longer gradually get higher priority.

Priority Scheduling Algorithm

Priority Scheduling result						
jobID	arrival	need	priority	start	wait	turnaround
6	700	10	1	700	0	10
1	800	50	0	800	0	50
3	830	25	2	850	20	45
4	835	20	2	875	40	60
5	845	15	2	895	50	65
2	815	30	1	910	95	125
7	820	5	0	940	120	125
total waiting time: 325						
total turnaround time: 480						
average waiting time: 46.43						
average turnaround time: 68.57						

Figure 4.2. Terminal showing the resulting output of HRRN

In priority scheduling, the program picked the arrived job with the highest priority. Here, a larger priority value was treated as a higher priority. The execution order was 6, 1, 3, 4, 5, 2 and 7. The total waiting time was 325 and the total turnaround time was 480. The average waiting time of 46.43 and an average turnaround time of 68.57.

This method focuses more on job importance than on arrival time or execution time. Therefore, lower priority jobs may have to wait much longer.

Combined Screenshot

```
hamidur@DESKTOP-7CGRQ8S:~/LUT_exercises/os_sp/ex3$ gcc -o job_schedule job_schedule.c
hamidur@DESKTOP-7CGRQ8S:~/LUT_exercises/os_sp/ex3$ ./job_schedule
please input job data file name
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Original job data
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jobID  arrival  need   priority
1      800     50     0
2      815     30     1
3      830     25     2
4      835     20     2
5      845     15     2
6      700     10     1
7      820     5      0

FCFS result
-----
jobID  arrival  need   start  wait  turnaround
6      700     10     700    0     10
1      800     50     800    0     50
2      815     30     850    35    65
7      820     5      880    60    65
3      830     25     885    55    80
4      835     20     910    75    95
5      845     15     930    85    100
total waiting time: 310
total turnaround time: 465
average waiting time: 44.29
average turnaround time: 66.43

SJF result
-----
jobID  arrival  need   start  wait  turnaround
6      700     10     700    0     10
1      800     50     800    0     50
7      820     5      850    30    35
5      845     15     855    10    25
4      835     20     870    35    55
3      830     25     890    60    85
2      815     30     915    100   130
total waiting time: 235
total turnaround time: 390
```

Figure 5. Terminal showing all the outputs together (part 1)

```

jobID  arrival need  start  wait  turnaround
6      700    10    700    0     10
1      800    50    800    0     50
7      820    5     850    30    35
5      845    15    855    10    25
4      835    20    870    35    55
3      830    25    890    60    85
2      815    30    915    100   130
total waiting time: 235
total turnaround time: 390
average waiting time: 33.57
average turnaround time: 55.71

HRRN result
-----
jobID  arrival need  start  wait  turnaround  response_ratio
6      700    10    700    0     10          1.00
1      800    50    800    0     50          1.00
7      820    5     850    30    35          7.00
2      815    30    855    40    70          2.33
5      845    15    885    40    55          3.67
4      835    20    900    65    85          4.25
3      830    25    920    90    115         4.60
total waiting time: 265
total turnaround time: 420
average waiting time: 37.86
average turnaround time: 60.00

Priority Scheduling result
-----
jobID  arrival need  priority  start  wait  turnaround
6      700    10      1      700    0     10
1      800    50      0      800    0     50
3      830    25      2      850    20    45
4      835    20      2      875    40    60
5      845    15      2      895    50    65
2      815    30      1      910    95    125
7      820    5       0      940   120   125
total waiting time: 325
total turnaround time: 480
average waiting time: 46.43
average turnaround time: 68.57
hamidur@DESKTOP-7CGRQ8S:~/LUT_exercises/os_sp/ex3$

```

Figure 6. Terminal showing all the outputs together (part 2)